Changelog

1.4.1

Follow-up to two field reports that turned out to be **version mismatch** - both symptoms were v1.2.1 behaviour, and both were already fixed in 1.4.0. The real finding was that nothing made that diagnosable, plus one genuine remaining hazard.

- `clear_document` **fallback sweep hardened.** The 1.4.0 rewrite used reverse-order `entity.deleteMe()` correctly, but kept a fallback that still bulk-deleted `bRepBodies` - the exact path that hung for 10+ minutes on a parametric document. That sweep now runs ONLY when there is no timeline (direct-modelling docs). On a parametric doc with leftovers it reports what remains and points at `retire_body()` instead of grinding. Verified: zero bulk deletes on a parametric doc.
- `ping_fusion` **now reports BOTH server and listener versions** and warns when they differ. The existing protocol handshake only caught a stale *listener*; a stale *server* (this case) produced no mismatch signal at all, so an old docstring and an old code path looked like live bugs.

1.4.0

Fixes from a full modelling session (splitter funnel). Several of these were silent-wrong-geometry bugs, which are the expensive kind.

Fixed (P0 - wrong facts that produced wrong geometry)

- **The documented YZ sketch-plane mapping was WRONG.** Docs claimed `sketch X -> world Y, sketch Y -> world Z`; measured behaviour is `sketch X -> world -Z, sketch Y -> world +Y`. Building off the wrong mapping yields a body with a correct bounding box and correct profile area that only reveals itself several operations later. Docstring corrected, and new injected helpers `P_xy/P_xz/P_yz` (world mm in, correct sketch point out) plus `verify_plane_mapping(sketch)` mean nobody has to reason about it.
- `report_volume_delta` **cried wolf on every Join.** It summed ALL bodies, so a Join that merges two bodies is volume-neutral and got flagged as a no-op - training you to ignore the one warning that matters. Now takes a `capture_state(target_body)` snapshot and warns only when body count AND volume are both unchanged.
- **Nothing detected silent geometry destruction by fillets.** A tangent-chained 1.5 mm fillet propagated ~20 mm and reverted a bore radius, reporting success with a valid body and no slivers.

`fillet_edges` now checks the volume delta against `~0.1*r^2*L` and warns on an order-of-magnitude overshoot; `fillet_volume_sanity()` exposes the same check to raw code.

Fixed (P1 - broken tools)

- **`export_stl` was unusable at every quality setting.** The enum names `*QualityTriangleMeshRefinement` do not exist on this build; the real members are `MeshRefinementLow/Medium/High/Custom`. Now uses those, with a graceful fallback that prints the available members if they change again.
- **`app.data` failures killed save/list/open for a whole session.** `save_document` now detects `InternalValidationError`, falls back to `doc.save()` for an already-saved document (the path that does work), and says clearly that a new lineage needs the Fusion UI. `list_recent_designs` explains the failure instead of returning a bare error.
- **`clear_document` hung for 10+ minutes and took Fusion down.** Reimplemented as reverse-order `item(i).entity.deleteMe()` (TimelineObject has no `deleteMe`) with `progress()` per feature, and it now runs through the async job system by default so the channel is never blocked.

Fixed (P2)

- **Bodies added via `BaseFeature` return a stale reference** - `ALL_TOOL_BODY_REFERENCE_LOST` when reused as a Combine tool. `cut_prisms` and `commit_tool_body()` now re-fetch by name after `finishEdit()`. This was a real bug in code shipped in 1.3.0.

Improved (P3 - observability)

- **`view_model`**: added `bottom`; an unknown orientation is now an ERROR instead of silently returning the previous view; added `zoom_x/y/z_mm` + `zoom_radius_mm` to frame a small detail instead of always fitting the whole model.
- **`cross_section`**: added `at_a_mm` / `at_b_mm` to pin the held axes, so you can scan through a specific feature rather than the bbox mid-plane.
- **`fillet_edges_smart`** - searches tangent-chain on/off x descending radii, verifies volume delta, rolls back overshoots, reports what worked. **`explain_fillet_error`** translates raw ACIS/ASM codes into fixes.
- **`smallest_faces(body)`** prints the actual smallest face areas; `sliver_face_count` threshold raised to 0.05 mm^2 (0.010-0.014 mm^2 faces are visible artifacts that the old cutoff missed).
- **Transport timeouts** now return a structured RUNNING status naming the job id, with an explicit "do not re-issue" note - a timed-out call had already completed a save in one case.
- **`retire_body()`** + docs for the replace-a-body idiom (`removeFeatures.add(old)`), which is strictly safer than clearing.

Docs

- New "API traps on this build" section in TOOLS.md covering the plane mapping table, `TimelineObject.entity`, stale BaseFeature handles, `is`-comparison on sketch entities, `addByCenterStartSweep` normalisation, `setMidRadii` list types, `app.data` failures, and silent fillet propagation.

1.3.0

Security fix plus the curved-surface toolset the docs already promised.

Fixed (P0)

- **SECURITY: argument injection.** Tool arguments were pasted directly into Python source text, so a filepath or name containing a quote broke out of the literal and executed arbitrary code inside Fusion. All ~22 interpolation sites now emit proper Python literals via `repr()`. Verified by executing generated code for 9 tools x 7 adversarial payloads (63 runs) with an injection canary: zero escapes.
- **P0: Windows paths raised SyntaxError.** Same root cause - `C:\parts\x.step` was pasted raw, so backslashes became escape sequences. Fixed by the same change; paths with backslashes, trailing separators and apostrophes now work.

Added (P1)

- **`wrap_cut`** - wrap a flat sketch onto a non-planar face and cut/emboss to depth. Previously referenced by the docs but MISSING from the build, so following the instructions led straight into a wall. Uses the correct `EmbossFeatures.createInput(profiles, faces, depth)` signature (negative depth cuts), cuts per-profile with individual volume checks, and reports ok/no-op/failed counts with sector guidance.
- **`project_to_surface`**, **`construction_plane_offset`**, **`construction_plane_tangent`**, **`sketch_axis_map`** - the supporting cast the wrap workflow needs. The construction-plane tools replace `ConstructionPlaneInput.setByPlane`, which is environment-blocked here, and echo world position so unit slips surface immediately.
- **`cut_prisms`** + injected **`tbm()`** / **`commit_tool_body()`** helpers - many prism cuts in ONE timeline feature via `TemporaryBRepManager`. Depends on neither construction planes nor face tempIds, so it sidesteps the two most fragile things in this environment.

Docs

- TOOLS.md gains a curved-surface section and a TemporaryBRepManager section with a worked example.

1.2.1

Fixes from a project-wide review triggered by a real degraded-state incident (old listener + new server = every payload empty while status read OK).

- **CRITICAL: switched to ThreadingHTTPServer.** The v1.2.0 listener used a single-threaded HTTPServer, so while `/exec_sync` held the one HTTP thread waiting on a job (up to ~110s), `/ping` and `/state` could not answer - silently defeating the "separate always-responsive channel" design. Verified fixed: ping now answers instantly during a blocked sync call.
- **Version/protocol handshake.** The listener now reports `listener_version` + `protocol` in every response; the server checks it and returns a loud PROTOCOL MISMATCH message with exact fix steps when the add-in is stale - instead of silently rendering empty payloads as STATUS: OK. This makes the incident that prompted this release impossible to miss.
- **Fail-loud capture health.** Envelopes include `stdout_bytes`; the server marks ok-but-silent results with "(no output captured)".
- **Job-store pruning.** Finished jobs are capped at the 50 most recent so long sessions don't grow memory unboundedly.
- **Docs:** INSTALL now warns that Fusion loads the add-in from its AddIns folder, not from wherever the project files live - updating one does not update the other.

1.2.0

Async execution and document safety — driven by a second round of real-use feedback. The core fix: a 4-minute timeout that told you nothing about what actually happened in Fusion. Execution is now decoupled from the HTTP response.

Added — async job model (the big one)

- **Listener is now async.** `POST /exec` enqueues a job and returns a `job_id` immediately; the operation runs on the main thread and is polled via `GET /job/<id>`, which returns a structured envelope (`status` , `stdout` , `traceback` , `elapsed_ms`) and **partial stdout while still running** (incremental progress).
- **`run_async(code)` + `job_status(job_id)`** tools for long operations that shouldn't block. Plus a `progress(msg)` helper to mark stages visible mid-run.
- **Ping/state on a separate channel.** `GET /ping` and `GET /state` are served from cache without the main thread, so they STAY RESPONSIVE during (or after) a hung operation. A hung op and a dead listener are now distinguishable.
- **`get_state()`** — cheap, guaranteed-fast document identity + counts (no geometry work). The safe first call for crash recovery.

- **Structured result envelope, always.** Every execution returns status/stdout/traceback/elapsed; no more bare "no result received". The sync path returns partial output and a job id if the op outlives the wait.
- **Watchdog** flags jobs that exceed their budget as `timeout` (clear signal rather than a dead channel).

Added — document safety

- **Document identity surfaced everywhere.** `get_state` and `get_model_state` lead with the active document **name + lineage id**, preventing wrong-document overwrites.
- `save_document(name, create_new=False)` now reports **which lineage it versioned**, and `create_new=True` saves a brand-new lineage so a new part never buries an unrelated design's history.
- `list_versions(design_id="")` — read-only enumeration of a lineage's saved versions, so a bad save is recoverable and the user can be pointed to the exact version to promote.
- `open_document` reports the new active lineage.

Added — documentation

- Sketch-plane / axis mapping (XY/XZ/YZ + offset planes, incl. the XZ→world-−Z inversion) documented in `run_fusion_code`'s docstring.

Notes

- `run_fusion_code` uses the bounded sync path (~100s) then hands off to polling; no behavior change for short ops.
- Volume-delta no-op detection and the `mm()` unit helper shipped in 1.1.0 and already cover two earlier feedback items.

1.1.0

Observability, diagnostics, and stability. Injected unit/observability helpers (`mm()`, `assert_scale()`, `report_volume_delta()`, etc.); auto volume-delta reporting with no-op warnings on cut/shell tools; `cross_section()`, `list_faces()`, `list_edges()`, `heartbeat()`; listener-level refusal of known host-crashing calls; richer `get_model_state()`.

1.0.0

First public release. Fusion 360 add-in listener + MCP server with ~30 tools (primitives, parameters, faces, assemblies/joints, measurement, visual feedback, export, persistence), preflight launcher, and full docs.